

AI Alignment as Normative Institutional Design

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Technical approaches to AI alignment have increasingly drawn on concepts developed for problems in moral philosophy, political theory and legal systems—from constitutions and social choice, to deliberation and virtue ethics. However, contemporary technical instantiations of these frameworks bring in these concepts unevenly. Some traditions (e.g., constitutions) have inspired substantial technical AI frameworks, while others remain largely unexplored. Moreover, alignment frameworks often retain only a fraction of the institutional structures developed in the source concept. In order to provide a systematic understanding of this landscape, we construct a two-dimensional taxonomy following the typology-construction of [Collier et al. \(2012\)](#) mapping normative AI alignment and governance regimes to theoretical concepts from philosophy or political or legal theory that they draw inspiration from. We use this framework to understand how normative concepts have so far been translated into technical systems; analyzing what is preserved, what is discarded, and when these omissions might have consequences. We highlight a recurring tendency of alignment methods to treat normative disagreement as a problem to be *resolved away*, despite governance frameworks that argue for not eliminating it. Lastly, by looking at the entire landscape, we identify missing components and underexplored regimes that could form future directions for building AI systems that can preserve, contest, and govern plurality rather than collapsing it.

Keywords: AI alignment, Governance, Philosophy, Political Theory, Legal Theory, Pluralism.

1. Introduction

What should an AI system do when the values it is expected to follow conflict? How should human preferences that differ across individuals determine its behavior? Should those preferences be aggregated into an objective, encoded as explicit principles, or resolved through deliberation? And when reasonable people continue to disagree, should an aligned system resolve it at all, or instead preserve and govern that disagreement over time?

These questions are increasingly central to AI alignment, but they are not uniquely questions about AI. Sub-fields of philosophy have long studied how competing values and obligations should be weighed ([Berlin, 1969](#); [Ross, 1930](#)); political theory has asked how institutions can legitimately govern populations that disagree about what is good or just ([Habermas, 1996](#); [Rawls, 1993](#)); and legal jurisprudence has developed systems for specifying, interpreting and contesting norms over time ([Dworkin, 1988](#); [Hart, 1961](#)). From these theories, real instantiations of con-

cepts have been implemented across institutions: from codified rules in constitutions to democratic procedures aggregating competing preferences; resulting in different institutional regimes for operating and governing under disagreement. In recent years, AI alignment has begun to draw from this repertoire of normative concepts. For example, constitutional AI ([Bai et al., 2022](#)) encodes normative principles in an explicit written artifact that trained AI systems are expected to follow; social-choice approaches to preference learning aggregate judgments according to individuals or incentives ([Ge et al., 2024](#); [Maura-Rivero et al., 2025](#)); deliberative approaches derive principles through collective procedures ([Huang et al., 2024](#)); and pluralistic systems attempt to allow model outputs to represent heterogeneous and sometimes incompatible values ([Feng et al., 2024](#); [Sorensen et al., 2025](#)). And as a result of the success of these approaches, such concepts originating in moral philosophy, jurisprudence and political theory have become a regular part of the technical vocabulary of alignment.

Yet this transfer of concepts is often selective.

For example, an alignment system inspired by constitutions (Bai et al., 2022) may borrow the codification of constitutions while implementing only a part of the entire framework—it preserves the idea of an authoritative set of principles written for a system, while omitting the institutional mechanisms through which these principles are interpreted, contested and revised, as they are in judicial systems. Such simplifications are often reasonable engineering choices: technical systems need not exactly reproduce their intellectual sources in every respect due to infrastructural or engineering constraints. However, our broader concern is that the **normative properties of a system can change when a concept is separated from the broader institutional structure in which that concept originally played a role**. We call attention to this phenomenon as a problem of *normative translation*. In this work, rather than asking whether technical implementations faithfully reproduce the theoretical concepts they originated from, we ask which features are preserved in translation, which are omitted, and what consequences follow from these choices.

[roma: todo rewrite and shorten below! keep introduction at 1.5 pages total?] In order to critically study this, we systematically map normative concepts to technical AI alignment systems, by asking what normative regime each method instantiates. Following Collier et al. (2012), we organize approaches according to two recurring attributes: where normative authority resides and what happens when normative claims conflict. Mapping existing approaches into this space reveals an uneven landscape: while alignment systems increasingly incorporate heterogeneous sources of normative input, comparatively few preserve disagreement as an enduring feature of the resulting system. This allows us to understand not only which normative concepts have influenced technical alignment systems, but also to understand both existing translations of normative institutions and regions of the design space that remain technically underexplored.

Our contribution here is threefold. First, we survey implementations of normative concepts in both alignment and governance regimes. We provide a searchable encyclopedia of these de-

finer concepts to aid researchers in work involving these concepts hosted at **[roma: xx]**. Second, we create a typological structure that organizes these concepts, and therefore the implementations under them, along dimensions of normative authority and conflict resolution. We discuss *coverage* of the space. Third, we document existing AI alignment systems that draw from normative concepts. We analyze when mistranslated concepts occur and potential consequences of this. Our goal is not to critique AI systems for not reproducing normative concepts identically, but rather, to enforce that borrowing of normative concepts from traditions should make institutional assumptions visible. By understanding the space of concepts holistically, and identifying the missing conceptual gaps in current alignment systems, we hope to identify both the strengths and limitations of existing approaches, as well as provide a systematic basis for exploring alternatives for future systems.

2. Related Work

Normative theory in AI alignment. A growing body of work draws explicitly on normative concepts from philosophy, political theory, and jurisprudence. Examples include value pluralism and incommensurability in pluralistic alignment (Sorensen et al., 2024), social choice for aggregating heterogeneous human preferences (?), deliberative democracy for collective alignment (Huang et al., 2024; Ovadya et al., 2024), constitutionalism for principle-based alignment (Bai et al., 2022), and jurisprudential concepts such as precedent and legal interpretation for case-based alignment and specification design (Caputo, 2024; Kolt et al., 2026). These works establish individual connections between normative traditions and technical mechanisms; we provide a broader mapping of these connections in Appendix ??.

Typologies and property spaces. Our approach builds on a longstanding tradition of typological analysis in political science and the social sciences, where multidimensional property spaces are used to form concepts, distinguish the-

oretically meaningful types, and expose combinations not captured by existing classifications (Barton, 1955; Collier et al., 2012; ?; ?). Rather than beginning with a set of observed categories, this tradition decomposes variation into underlying dimensions whose cross-product defines a property space; importantly, theoretically possible but empirically unoccupied cells can themselves be analytically informative. We adopt this approach to construct a typology of normative alignment architectures from two recurring dimensions: where normative authority resides and how normative conflict is treated.

Our approach. Recent AI taxonomies instead primarily organize systems according to technical mechanisms or particular forms of value pluralism. For example, Sorensen et al. (2024) distinguish Overton, steerable, and distributional pluralism, while other frameworks distinguish approaches according to how preferences or perspectives are represented and aggregated. We instead construct a *property space* (Collier et al., 2012) from recurring attributes of normative systems. Rather than grouping approaches according to their surface-level technical mechanisms or intellectual origins, this representation allows methods from different traditions to be compared according to where normative authority resides and how normative conflict is treated. Importantly, the property-space approach also retains unrealized combinations, allowing gaps in the technical literature to become objects of analysis rather than being excluded from the taxonomy.

3. A Normative Design Space for Alignment: A Taxonomy

[roma: this needs to be the biggest section. we need to clarify how the taxonomy was created/why this current form is the best]

In order to study the use of normative concepts across fields, we create a two-dimensional property space defined by two axes: the locus of normative authority and the disposition of normative conflict. We derive these dimensions through a systematic mapping of alignment and governance literature followed by conceptual substructure

(Barton, 1955): "the procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937). Overlapping surface categories are therefore decomposed into the underlying attributes that distinguish them. Following Petersen et al. (2008), two coders independently extract normative concepts from the sampled literature, reconcile near-equivalent categories and extract keywords that represent the respective concept. Keywords are then unified across studies, with near matches merged together, where the resulting clusters form categories of the scheme¹ Unlike conventional typologies that discard unrealized combinations, we retain empty cells because absence from the technical literature is itself analytically informative for identifying underexplored architectures.

This yields two dimensions: the **locus of normative authority**, which indicates where the binding normative content resides prior to the moment of decision, and **disposition of normative conflict**, which indicates how any conflicts should be resolved. Their cross product gives a 5 x 3 property space. For the locus of authority axis, we find the following categories:

1. **Codified.** Norms to be followed exist as explicit written specifications, that already set, and in in principle, complete before any particular case arises.
2. **Case-accumulated.** Norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance; but by what accumulated decisions have held (Jonson and Toulmin, 1988; Levi, 2013).
3. **Customary.** Norms reside in regularities of practice, never written down and never adjudicated. There is no authoritative moment at which a norm gets explicitly stated (Burke, 1986; Lewis, 2008).
4. **Dispositional.** The norms reside in the individual's character rather than in any external artifact. There is no rule to consult and nothing to look up, only evidence of traits of the individual itself (Aristotle).

¹Full collection and inclusion criteria appear in Appendix A.

5. **Procedural.** Norms persist between decisions. What should be done is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (Habermas, 2015).

For the disposition of normative conflict, we find the following categories:

1. **Collapse.** When norms conflict, the system resolves the conflict into a single output, without trace or acknowledgment of the disagreement that produced it.
2. **Contest.** The conflict between norms is preserved and kept procedurally live. Either the output itself carries the disagreement, or the decision stays open to challenge through a standing channel (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is neither resolved nor sustained but routed around. Multiple normative orders are kept intact, and a rule assigns which one governs a given case (de Sousa Santos, 2002; Lijphart, 1977; Savigny, 1867).

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4. Case Studies in the Design Space

[roma: this can cover 2-3 case studies in about 1 page.] Here, we use our typology to examine representative alignment paradigms, to ask the question: **how have normative concepts been translated into technical systems and what assumptions are made?** Our purpose is not just to evaluate the fidelity of technical implementations to their philosophical sources, but to examine what functions of the original normative framework are preserved, as well as how design choices affect the resulting governance structure.

²We considered open, axial and selective coding (Corbin and Strauss, 2014), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), but note that the keyword-based approach is more native to the computer science literature it addresses. Alternatives would likely yield a comparable outcome in terms of the intermediate categories.

4.1. Preference-based Alignment

Preference-based alignment of LLMs is a widely used paradigm that involves training a reward model to infer human preferences over LLM outputs, and then using this model within the RL training loop to steer LLM outputs towards the approximated aggregate preferences. The resulting model is meant to be *aligned* to users as a result of this optimization towards the aggregate of all the user's preferences. This idea has roots in utilitarian moral philosophy—Bentham's principle of utility Bentham (1994) evaluates decisions according to their effect on the aggregate happiness of humans affects, while Mill posits that individual welfare contributes to an overall social good Mill (2011).

The many critiques against this include Tocqueville and Mill that warn against the *tyranny of the majority* i.e., a resulting state where aggregate or majoritarian consensus can erase or suppress minority perspectives and contradicting world views. Later work in social choice theory and welfare economics provide theoretical arguments against the validity of these frameworks showing how no rank-order voting systems—such as those used in preference algorithms—can convert individual preferences into a population-level consensus while maintaining fairness axioms². Although preference-based alignment aims to elicit comparative judgements from humans at data collection time, they ultimately contribute to a common reward function, rank ordering, or model policy—thus only representing the aggregate population vote. Technical translations therefore make an architectural choice to represent plurality upstream but then collapse it downstream. In our taxonomy, this aggregation falls directly under *collapse*: where competing judgements are transformed into a common ordering. [roma: spend at least 2-3 sentences talking about disagreement work, but how it hasn't made it into mainstream alignment techniques] There do exist alternative architectures in which disagreement survives, and where minority judgements or conflicting normative claims can remain represented downstream. There is a big design opportunity therefore for systems that treat disagreement as information to gain signal from rather than noise

to be eliminated.

4.2. Constitutions and Rules

Moving away from utilitarian preference aggregation, Constitutional AI locates normative authority primarily in a codified artifact: an explicit set of principles specified prior to a particular model decision. Yet a constitution in this technical sense performs only some of the functions associated with constitutions in legal and political theory. Constitutional governance ordinarily concerns not only the existence of a written text, but also questions about who may interpret it, how conflicts among provisions are resolved, how its application develops through precedent, under what conditions it may be amended, and which institutions possess authority to contest or revise decisions. A fixed specification largely separates the normative text from this surrounding institutional machinery.

This translation is not necessarily a defect. Reducing a constitution to an explicit specification makes normative guidance legible, auditable, and usable within existing training pipelines. But it also changes the normative architecture. Authority that is distributed across text, interpretation, precedent, and institutional roles becomes concentrated in the specification itself and in the process that authors it. Disagreement that constitutional institutions may keep procedurally live can instead be resolved before deployment, when principles are selected and encoded.

This comparison therefore suggests a different question from whether Constitutional AI is “sufficiently constitutional”: which functions of constitutional governance are useful for alignment? Existing work has emphasized the specification of principles. Less explored possibilities include mechanisms for authoritative interpretation, amendment, appeal, precedent, or explicit resolution of conflicts among constitutional principles. Normative translation makes these omissions visible as design choices rather than treating “a constitution” as a unitary technical object.

A related omission concerns the *distribution of institutional roles*. Normative institutions typically separate, to varying degrees, the actors who

specify rules, implement them, evaluate compliance, and adjudicate failures. AI alignment often concentrates these functions within the model developer. The 2026 OpenAI–Hugging Face incident illustrates why this distinction can matter: following an internal evaluation in which model behavior escaped its intended testing environment and affected third-party infrastructure, OpenAI commissioned METR and Redwood Research to conduct an independent investigation (??). Subsequent policy proposals have similarly emphasized independent verification and third-party auditing of frontier-model safety claims. This suggests another dimension of normative translation beyond the content of rules themselves: *who is institutionally authorized to specify, interpret, evaluate, and contest them*.

4.3. Deliberation and Collective Judgements

Deliberative approaches locate normative authority in a procedure through which participants exchange reasons and construct or revise judgments collectively. Their normative analogues, however, depend not merely on communication but on institutional conditions governing participation, justification, challenge, and legitimacy. Technical systems can therefore reproduce deliberation’s surface mechanism while omitting these surrounding procedural guarantees.

Importantly, regarding our second dimension, deliberation does not uniquely determine what happens to disagreement. A deliberative procedure may seek convergence on a single principle or policy, in which case it remains a collapse architecture. Alternatively, it may preserve dissent, minority reports, or opportunities for subsequent challenge, moving toward contest. Thus, labeling a system “deliberative” is insufficient to characterize its normative architecture: the same procedural mechanism can instantiate different treatments of normative conflict. This illustrates why separating our two dimensions is useful. Where normative authority comes from does not determine what a system must do with disagreement. Procedural systems need not collapse, just as codified systems need not be uncontestable. Normative theory makes these combinations visible even where current technical implementations

concentrate on only a subset of them.

5. Expanding Design Space

[roma: this can discuss what is missing and future gaps to fill] Across these examples, technical translation tends to preserve the component of a normative institution that can most readily be represented as a computational mechanism: constitutions become specifications, heterogeneous preferences become aggregation problems, and deliberation becomes a procedure for producing judgments. The surrounding mechanisms through which normative institutions manage interpretation, disagreement, revision, and competing authority are less consistently retained.

We do not interpret this pattern as evidence that existing alignment methods are incorrect implementations of their theoretical analogues. Computational systems face different constraints and objectives from legal or political institutions, and simplification is often necessary. Rather, the comparison exposes where normative decisions have moved. A fixed constitution moves questions of interpretation and conflict toward specification design; an aggregate preference model moves disagreement into the aggregation rule; and consensus-oriented deliberation moves conflict into the procedure that determines its outcome.

Contest Existing red-teaming already institutionalizes adversarial disagreement, but primarily as an epistemic instrument: opposing roles discover failures that are subsequently folded into a unified model or policy. Contest-oriented alignment could instead preserve adversarial roles within the normative decision process itself, allowing decisions to be challenged and successful challenges to alter subsequent behavior.

Partition TODO choice-of-law, jurisdiction-specific constitutions.

This observation also connects normative translation to the gaps in our property space. Components omitted during translation frequently resemble the mechanisms that distinguish contest

and partition architectures: appeal, amendment, persistent dissent, jurisdiction, competing sources of authority, and procedures for revisiting prior decisions. The sparsity of these regions may therefore reflect not an absence of normative resources for designing such systems, but the selective translation of those resources into alignment to date.

The value of comparison with normative theory is consequently generative rather than merely diagnostic. It allows us to ask not whether alignment systems faithfully reproduce existing institutions, but which institutional mechanisms become available once we stop treating their current technical translations as exhaustive. We turn next to these underexplored architectures and the design opportunities they suggest

Our analysis reveals two sources of underexplored alignment designs. Architectural gaps arise where few or no existing methods occupy a region of the property space, suggesting combinations of normative authority and conflict management that have received limited technical attention. Mechanism gaps arise within existing architectures when potentially useful components of their normative analogues—such as contestation, precedent, amendment, or jurisdiction—are absent from technical implementations. Neither type of gap implies that the missing design should be adopted: some may be undesirable or impractical for AI systems. Rather, they provide structured hypotheses about alternative alignment mechanisms whose tradeoffs can be implemented and evaluated.

6. Conclusions

[roma: also either bleed in limitations here, or have a section for it before this]

AI alignment is not only a problem of identifying which values a model should represent, but also of deciding how normative authority should be structured when those values conflict. We introduced a taxonomy that makes these choices explicit, organizing alignment approaches by where normative authority resides and how normative conflict is treated. Viewing existing alignment systems through this lens also makes visible the

consequences of *normative translation*: technical implementations preserve some features of the philosophical, political, and legal traditions they draw from while leaving others behind. Making these design choices explicit expands the space of possible alignment approaches, particularly toward systems that can preserve, contest, or partition persistent disagreement rather than requiring it to collapse into a single normative objective.

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Acknowledgements

7. Appendix